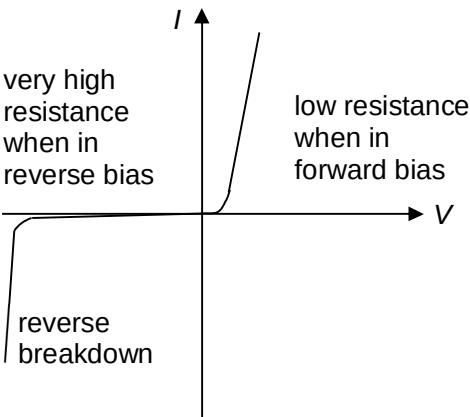
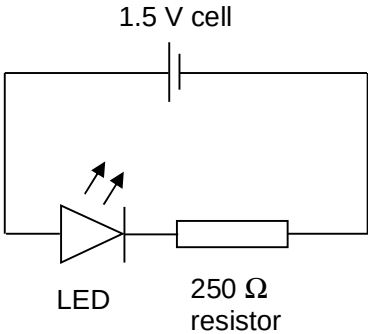


	carbon-14, which is 5700 years.
7(a)	The resistance of a circuit component is the ratio of the potential difference across the component to the current in it.

7(b)(i)	From Fig. 7.1, - length of XY = 6.3 cm - length of ZY = 4.4 cm $V_{YZ} = \left(\frac{4.4}{6.3} \right) (5.0) = 3.5 \text{ V}$
7(b)(ii)	$R = \frac{\rho L}{A}$ where A is the cross-sectional area of the wire $R = \frac{\rho L}{A} = \frac{\rho L}{\frac{m}{DL}} = \frac{\rho DL^2}{m}$
7(b)(iii)	If the length of the wire is doubled, $R' = \frac{\rho D(2L)^2}{2m} = 2R$ OR $R' = \frac{\rho(2L)}{A} = 2R$ Therefore the resistance doubles. The student is wrong.
7(c)(i)	 Comments in the Examiner's Report <i>This question was poorly answered. The majority of candidates did not realise that <u>in reverse bias a diode does not conduct</u> and showed a non-zero current. Many candidates did not <u>show a threshold p.d. below which the current is zero and above which the current rises rapidly</u>. Many gentle curves from zero were shown. Some candidates showed the same incorrect graph with a slight positive gradient of the line running in third quadrant and into the first.</i> The first underlined statement indicates an ideal semiconductor diode in reverse bias. The second underlined statement indicates a non-ideal semiconductor diode in forward bias.
7(c)(ii)	Very high resistance when connected in reverse bias. Infinite resistance when connected in forward bias and below threshold p.d. Low resistance when connected in forward bias and above threshold p.d.

7(c)(iii)	 1.5 V cell LED 250 Ω resistor
7(c)(iv)	The resistor is necessary to limit the current and prevent damage to the LED.
7(d)(i)	$V = \left(\frac{1.4}{1.4 + 1.5} \right) (12) = 5.8 \text{ V}$
7(d)(ii)	As the temperature of the resistor rises, the resistance of the thermistor decreases. The effective resistance of the circuit decreases. The current in the circuit increases. The p.d. across the fixed resistor increases.
8(a)	They will hit the ground at the same time as they have the same initial vertical